

EXCAVATOR SIMULATOR: SEMG-BASED HMI TECHNOLOGY USING DEEP LEARNING TECHNIQUES



Final Year Project Report

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June 2024

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A Dissertation
Presented to
The Academic Faculty

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In Partial Fulfillment
of the Requirements for the Degree of
BS ELECTRICAL ENGINEERING in the
INFORMATION TECHNOLOGY UNIVERSITY OF THE PUNJAB
LAHORE, PAKISTAN

Department of Electrical Engineering
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Lahore, Pakistan
Year 2024

CERTIFICATE OF ORIGINALITY

We hereby declare that this report titled “Excavator Simulator: sEMG-Based HMI Technology Using Deep Learning Techniques” is our own work to the best of our knowledge. It contains no materials previously published or written by another person, nor material which to a substantial extent has been accepted for the award of any degree or diploma at ITU or any other education institute, except where due acknowledgement, is made in the thesis. Any contribution made to the research by others, with whom we have worked at ITU or elsewhere, is explicitly acknowledged in the thesis.

We also declare that the intellectual content of this report is the product of my own work, except to the extent that assistance from others in the project’s design and conception or in style, presentation and linguistics is acknowledged. We also verified the originality of contents through plagiarism software.



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ACKNOWLEDGMENTS

We want to express our sincere gratitude to Dr. Tahira Mahboob, our FYP advisor, and Dr. Rehan Hafiz, our co-advisor. Their unwavering guidance and support were instrumental in helping us complete our FYP successfully. Their expertise and mentorship were invaluable, and we are truly thankful for their contributions to our academic journey. We would also like to acknowledge that our project was funded by the Pakistan Engineering Council (PEC), which greatly facilitated our research and development efforts.

DEDICATION

This dissertation is dedicated to our parents.

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Abstract

Pakistan's emerging VR industry faces significant challenges due to the high costs of imported equipment, prompting our initiative to develop a budget-friendly alternative. Our project focuses on wearable surface electromyography (sEMG)-based technology as a cost-effective substitute for expensive imports like Haptic gloves and Myo band arms. Emphasizing the development of a Human Machine Interface (HMI) for an immersive Metaverse experience, our vision materializes in an sEMG-based excavator simulator. The current research gap lies in the lack of affordable yet effective VR training solutions for industrial applications, specifically in the area of machine operator training. Traditional VR equipment is prohibitively expensive, limiting its accessibility and widespread use. Our research aims to bridge this gap by developing a cost-efficient, high-performance sEMG-based simulator that replicates real excavator operations. Our innovative approach captures muscle signals through sEMG technology and incorporates Force-Sensing Resistors (FSRs) to detect the pressure exerted by users during virtual excavator operations. By strategically placing EMG sensors and integrating FSRs, our simulator offers an ultra-realistic training experience, ensuring that every nuance of real excavator operations is mirrored accurately. In terms of performance, the proposed excavator simulator is a revolutionary product designed to target industrial applications for training and practice by machine operators and trainers. Our results indicate that the sEMG-based simulator is not only cost-efficient and time-conservative but also a safe solution for providing an immersive training experience. This groundbreaking solution significantly reduces training costs and enhances operator skills, thereby revolutionizing VR training and immersive experiences in the industrial sector.

CHAPTER 1

INTRODUCTION

1.1 Motivation

We're developing a specialized training tool for heavy machinery operators with several key benefits. Safer and easier instruction in the operation of large machinery is our goal. We are working on a project called EXCAVATOR SIMULATOR, which is a virtual [1]excavator controlled by advanced sensors-based technology. The sensors help to prevent mishaps by monitoring the operator's movements and even detecting when they are becoming too weary [2]. Our primary objective is safety. Saving money and protecting the environment are important priorities for us. We save energy and maintenance costs by using our simulator instead of actual machinery. In addition to lowering the cost of training, this lowers emissions and benefits the environment.

1.2 Challenges

Developing a simulator for excavator operators that uses special sensors to read muscle signals can be challenging. Getting accurate and sufficient muscle signal data is crucial [3], and making the simulator feel realistic like operating a real excavator is a complex task. Teaching computer systems to quickly understand and respond to muscle signals is a difficult aspect of the technology [4]. Additionally, integrating both surface electromyography (sEMG) sensors and Force-Sensing Resistors (FSRs) to work seamlessly together, and processing the data from both sensors accurately, presents significant challenges. Adapting the system to work effectively for different users and handling variations in muscle signals due to fatigue are important considerations. Providing effective feedback to users for skill improvement, managing costs, and complying with industry regulations are all essential factors. Maintaining the simulator and ensuring it remains user-friendly are also key aspects of the development process.

1.3 Sustainability

Modern and environmentally sustainable, the Excavator Simulator uses muscle signal technology. This simulator functions virtually, reducing its environmental impact, in place of actual machinery that may contribute to pollution. Another significant benefit is its affordability, as it eliminates the need for fuel and repairs that come with actual machinery. Long-term cost reductions result from this, which also helps the environment and operators' jobs. The range of possibilities is what sets this simulator apart. Each person's experience is customized, enabling ongoing development and individualized training. It is essential for availability in addition to being cost-effective. This technology allows people to learn and develop their skills even if they do not have access to actual devices.

1.4 Problem Statement

Our project tackles the challenge of creating a realistic excavator simulator using EMG-based technology and deep learning techniques. The main problems include obtaining accurate EMG data, ensuring real-time model precision, and adapting to user variations. We must also consider cost-effectiveness, environmental sustainability, regulatory compliance, and user-friendly integration with real excavator hardware. Solving these challenges is vital for an effective and accessible training tool for excavator operators.

1.4.1 Unmet Need

We need an Excavator Simulator with EMG-based HMI Technology using Deep Techniques because we have important problems to solve. We want to make things safer, save money, protect the environment, help people learn better, and follow the rules. This kind of simulator can help with all these challenges in the construction and heavy machinery industry. We can make this simulator more affordable and accessible through low-cost and Indigenous development.

1.4.2 Potential Market

The potential market for an Excavator Simulator using EMG-based HMI Technology with Deep Learning is significant. It includes construction and heavy machinery industries, training institutions, equipment manufacturers, safety agencies, environmental initiatives, and a global market. This technology offers valuable operator training, safety, and environmental sustainability solutions, making it relevant to various sectors.

1.5 Report Outlines

The main scope of this report is the following:

Chapter 1 describes the introduction of the project including the motivation, challenges, sustainability, problem statement, unmet need, and the potential market of our project.

Chapter 2 comprises the literature review including the basis of EMG signals and force sensing resistor pad.

Chapter 3 proposed methodology and architecture include the proposed solution, packing, hardware components, and software components.

Chapter 4 consists of milestones, work division, and cost.

Chapter 5 concludes the complete project.

CHAPTER 2

LITERATURE REVIEW

In recent years, significant progress has been made in the fields of excavator simulation and virtual reality technology [5]. One notable development is the creation of an open-source PC-based excavator simulator, powered by the Delta3D engine. This simulator has greatly improved training and the evaluation of control strategies, offering a more realistic and precise learning environment. Simultaneously, the introduction of an EMG-based human-machine interface for VR environments has been remarkable . This innovative HMI employs lightweight forearm sEMG sensors [6] and a neural network model to decode finger forces in real-time, significantly enhancing users' perception of virtual object properties and making control interactions more intuitive. These advancements hold substantial potential for elevating the quality of training and user experiences in both excavator simulation [7] and VR applications, catering to diverse user demographics.

Ensuring accessibility and inclusivity within these technologies remains a paramount concern to expand their user base and accommodate individuals with varying abilities. Furthermore, using simulators and VR for training can help reduce pollution and save energy compared to real-world training, which is a significant environmental benefit. Lastly, hybrid training approaches, integrating physical equipment with virtual environments, present a comprehensive learning paradigm that seamlessly combines simulation with real-world training for excavator operators . Improvements in computer interactions can be seen in the literature concerning pressure-sensing devices such as UnMousePad [8]. In order to register touches and identify actions like touch-ups or touch-moves, researchers are developing intelligent algorithms that transform raw pressure data into meaningful information [9]. In order to make the data more useful, they are also refining the processing method. These gadgets could find application in areas other than computers, such as gaming controllers or clothing. Future research aims to incorporate touch reactions and further enhance the intelligence of these devices, hence increasing their use for all users.

In the realm of prosthetics and rehabilitation engineering, combining electromyography (EMG) technology with virtual robotics is showing promise. This review looks at a virtual robotic arm controlled by EMG signals using the Myo armband and Unity 3D [10]. The distinctive feature of this system lies in its adeptness at detecting electrical signals in forearm muscles, enabling the precise control of a virtual robotic arm without the need for dynamic motions. This is especially beneficial for hand amputees, addressing challenges in replicating fine motor control. Experiments confirm the system's effectiveness in interpreting EMG gestures, suggesting potential use in rehabilitation and assisting those with hand amputations. This innovative approach represents a significant step forward, offering improved quality of life for individuals with limb loss through its intuitive control mechanisms.

Force-sensing resistors (FSRs) are used as a component of its experimental setup to complement Electromyography (EMG) sensors in capturing data related to hand-object interactions. FSRs are utilized to measure finger-wise force data alongside EMG sensors, contributing to a comprehensive understanding of the biomechanical aspects of hand interactions. FSRs are employed to provide additional information about the force exerted by individual fingers during various hand-object interactions. Integration aims to improve data accuracy and completeness, offering insights into the relationship between muscle activity and hand force during virtual and augmented reality interactions. FSRs, alongside EMG sensors, enable a thorough investigation into force dynamics in hand interactions. Overall, FSRs are utilized alongside EMG sensors to provide detailed insights into the biomechanics of hand interactions [11], enriching the dataset and enabling a comprehensive investigation into force dynamics during virtual and augmented reality interactions.

Previous projects have utilized various devices to capture muscle signals, such as the Myo armband and Google band, which are quite expensive. In contrast, our project aims to reduce costs by employing more affordable EMG sensors and electrodes to capture muscle signals [12]. This substitution significantly lowers the overall cost compared to using expensive armbands. Additionally, unlike earlier works that did not incorporate Force-Sensing Resistors (FSRs), our project integrates FSR pads to detect the pressure applied by users. This allows our simulator to adjust the depth of excavation based on the amount of pressure applied—digging deeper with more pressure and shallower with less [13]. This innovative approach not only makes our project more cost-effective but also enhances the realism and functionality of the training experience. Excavator accidents are a significant concern in various industries due to their potential to cause

injuries, fatalities, and property damage [14]. According to Figure 2.1, there was an average of 143 deaths annually. These incidents are often attributed to factors such as operator errors, insufficient training, mechanical failures, and unsafe working conditions. Our prevention measures will include enhanced operator training, reduce pollution, and save energy compared to real-world training.

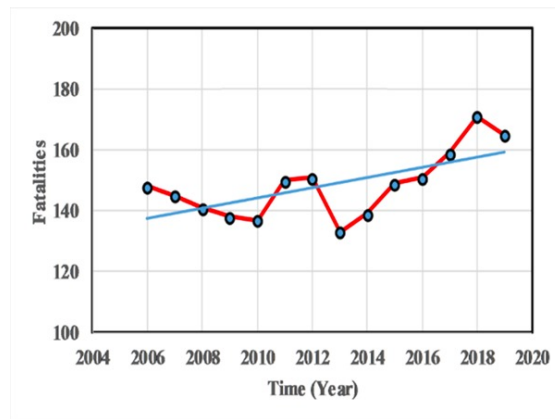


Figure 2.1: Trend of mining accidents in Pakistan from 2006 to 2019

[14]

2.1 Overview of Research Findings

Year	Paper Name	Findings	Research Gaps
2017	Hand Rehabilitation Using Virtual Reality and Electromyography Signals [2].	VR combined with EMG signals in this study proves effective for stroke rehab. An EMG-VR system translates muscle activity into virtual hand movements for engaging exercises.	This EMG-VR system needs further development to improve gesture recognition accuracy. Additionally, the research suggests incorporating more sensors and exploring the system's use in broader rehabilitation areas.

2018	EMG-based gesture recognition using machine learning [15]	This study explores using EMG signals and machine learning for gesture recognition. Their system using DWT and SVM achieved 83% accuracy in recognizing hand gestures.	This EMG gesture recognition system could be improved by exploring different algorithms or feature extraction methods to boost accuracy beyond 83%.
2019	An EMG-Based Virtual Reality Application for Motor Rehabilitation [16]	EMG-controlled VR game in this study shows promise for motor rehab, with users performing 99% better than traditional exercises.	The study needs testing with real rehab patients and exploring more VR scenarios to keep users engaged over time.
2019	The UnMousePad: an interpolating multi-touch force-sensing input pad [8]	FSR tech enables precise pressure sensing in a flexible, multi-touch device called the UnMousePad. This opens doors for innovative touch input applications, but further development is needed.	UnMousePad requires further exploration of software integration and development in transparent forms and diverse form factors.
2020	Emg-based decoding of manipulation motions in virtual reality Towards immersive interfaces [7]	This study proposes an EMG system for VR that decodes hand motions during object manipulation. The system achieves high correlation (92%) and good accuracy (83%) in translating muscle signals to virtual object movements	This EMG-VR system, while accurate, could benefit from adding haptic feedback to enhance user experience and suggests exploring real-time applications where the system.

2021	Reflections on using a game engine to develop a virtual training system for construction excavator operators [17]	This paper shows VR's potential for safe and cost-effective construction excavator training using Unity3D but highlights the need for more research on its effectiveness.	While the paper showcases a VR excavator simulator, further research is needed to understand how VR training translates to real-world skill acquisition.
2022	Force-Aware Interface via Electromyography for Natural VR/AR Interaction [9]	This study shows an EMG system can predict finger forces in VR/AR with high accuracy. The predicted forces allow for more natural interactions with virtual objects compared to position-based methods.	This EMG system needs testing to see if it generalizes well to different users beyond the single user in the study. Additionally, research is needed to explore how the system performs in a full VR/AR environment for real-world usability testing.
2023	EMG-based 3D hand gesture prediction using transformer-encoder classification [18]	This study combines sEMG signals with optical tracking for 3D hand gesture prediction using a Transformer-Encoder Classifier. The system achieves high accuracy (up to 98.6%) in predicting six hand gestures.	While achieving high accuracy, this TEC system requires further exploration of how its components impact performance. Additionally, the research suggests expanding the system's ability to recognize more than just six gestures.
2024	Our Work	Our project work created an indigenous development, cost-effective sEMG-based excavator simulator using EMG sensor's and pressure sensors, offering realistic and accurate training for operators.	Future improvements should include using more muscle sensors for better accuracy and exploring advanced data processing to enhance real-time signal interpretation and user adaptability.

Table 2.1: Summary of Research Papers

CHAPTER 3

PROPOSED METHODOLOGY AND ARCHITECTURE

3.1 Engineering Design

The following steps were used in the Engineering Design process to solve the problem.

- Ask: With a rising number of accidents, Pakistan faces challenges in providing optimal training for excavator operation, particularly in construction sites, due to resource constraints. While addressing this issue, certain constraints and limitations in terms of features may arise as a result of budgetary limitations.
- Research the problem : Training a machinery operator is a laborious and costly task, as it requires the support of heavy machinery, instructor, time, etc. Many construction companies cannot give the proper training to operators and they do have not the proper setup to give the training. Hence people living there are deprived of these resources.
- Imagine: Meticulously research, analyze, investigate the problem under consideration and propose a viable and feasible solution for the target market. Develop possible solutions brainstorm ideas and come up with as many solutions for this problem as possible.
- Plan: Develop low-cost EMG-BASED HMI technology using deep learning techniques which will give the training to the operators, which were based on trained data sets through Deep Learning.
- Create: The initial prototype was a single EMG sensor module and FSR pad in which we will find the muscle signal wave.
- Test and evaluate prototype: We first tested this prototype on a few selected subjects to detect whether they showed hand movement by applying an EMG sensor on the arm, also applying pressure on the FSR pad and showing output in the form of a wave.
- Improve: Redesign as needed

There were some wrong detections of the EMG sensor module mainly due to fewer data sets for training the model. To fix this we decided to get more and more data for accuracy. After achieving this we will further improve our design so that it can detect more accurate waves and add an FSR pad e.g. upward hand (compress wave), downward hand (relax wave), no pressure, light pressure and hard pressure.

3.2 Objectives and Outcomes

- **Develop Cost-Efficient Hardware**

Objective: Design and manufacture affordable sEMG sensors and FSR-integrated hardware.

Outcome: Affordable sEMG-based wearable device Integration of FSRs for pressure detection

- **Create High-Performance Software**

Objective: Develop software to accurately interpret muscle signals and simulate excavator operations.

Outcome: Software that captures and processes sEMG data Realistic simulation environment for excavator operations

- **Deliver Effective Training Results**

Objective: Provide a realistic and immersive training experience for machine operators.

Outcome: Enhanced training effectiveness Reduced training costs Improved operator skills and safety

- **Utilization of Open Source Tools**

Open-Source Hardware Designs: To reduce development costs and encourage community collaboration.

Open-Source Software Libraries: For signal processing, machine learning, and simulation.

Community Forums and Collaborative Platforms: For continuous feedback and improvement.

Outcomes: Using open-source Unity 'unity software link' with an excavator avatar 'excavator avatar download link' plus EMG sensor and FSR hardware 'source'.

3.3 Proposed Solution

After conducting an in-depth literature review for our proposed project, we have observed that most of the solutions available in the market rely on the use of hand-sensing gloves. However, these gloves pose challenges for users, as they can be uncomfortable and rigid, making them unsuitable for extended periods of wear. Another approach involves the utilization of expensive arm bands, rendering them inaccessible to a wide range of potential users due to their high cost. The primary drawbacks of these solutions are the discomfort associated with gloves and the high expense of armbands. In response to these limitations, we have developed a novel solution based on an EMG biosensor module. This innovative approach entails the measurement and interpretation of muscle movements in the human body using strategically placed electrodes. This enables us to accurately capture hand movements without the need for uncomfortable gloves or costly armbands. Our approach not only eliminates the inconveniences associated with traditional solutions but also provides a cost-effective alternative, ensuring accessibility to a broader user base. The ease of use and affordability of our EMG biosensor module makes it a promising and practical solution for a wide range of applications in the field of motion sensing. Figure 3.1 shows the Block diagram of the Proposed Solution. The Figure 3.2 shows the EMG and FSR pad data collection prototype.

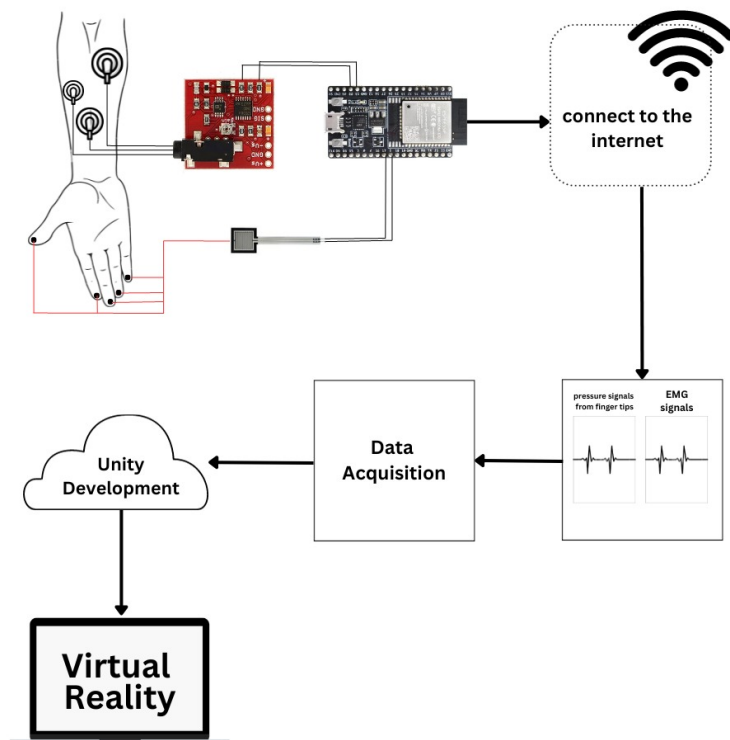


Figure 3.1: Block diagram of Proposed Solution.

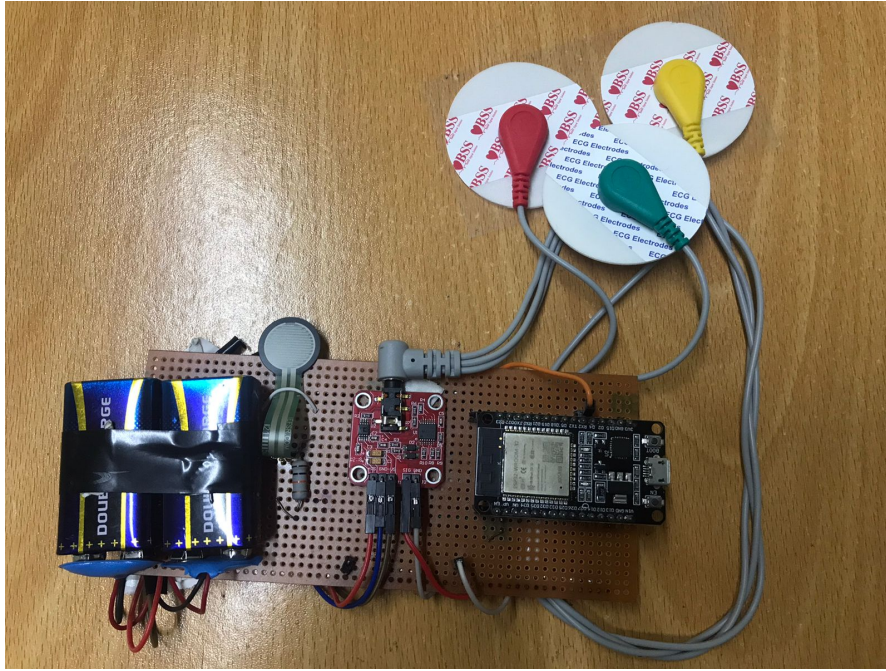
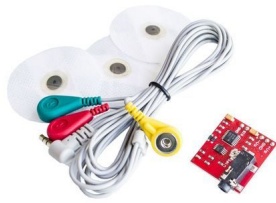


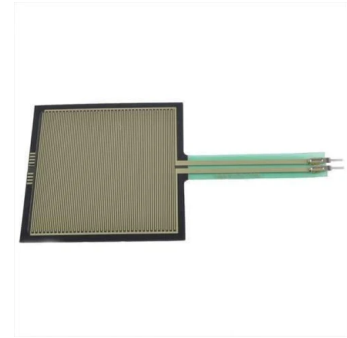
Figure 3.2: EMG and FSR pad data collection prototype.

3.4 Hardware Components

The hardware setup includes an EMG biosensor module, illustrated in Figure 3.3(a), which operates within a voltage range of 3.5V to 9V DC. Figure 3.3(a) also depicts the presence of three electrodes, which serve the purpose of acquiring voltage signals from arm muscles using a lead configuration. Additionally, there is a force-sensitive resistive pad, as shown in Figure 3.3(b), which detects and measures the pressure applied by the finger on the board, indicating whether the pressure is low or high. The microcontroller board that we are using is the ESP32 developer version, depicted in Figure 3.4(a). This microcontroller has multiple I/O pins and two analog channels, enhancing its versatility for our project needs. Furthermore, it comes equipped with integrated WiFi and Bluetooth modules, enabling seamless transmission of our sensor readings to unity. To power, both the ESP32 board and the sensors utilized in our module, a dedicated battery pack is incorporated into the setup, as shown in Figure 3.4(b). In Figure 3.5, there is the hardware of our project.

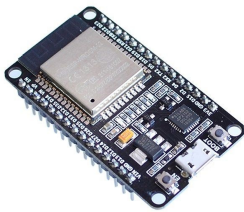


(a)



(b)

Figure 3.3: Components (a) EMG Biosensor Module Kit 'link', (b) Force Sensitive Resistive Pad 'link'.



(a)



(b)

Figure 3.4: Components (a) Microcontroller ESP32 'link', (b) Battery 'link'

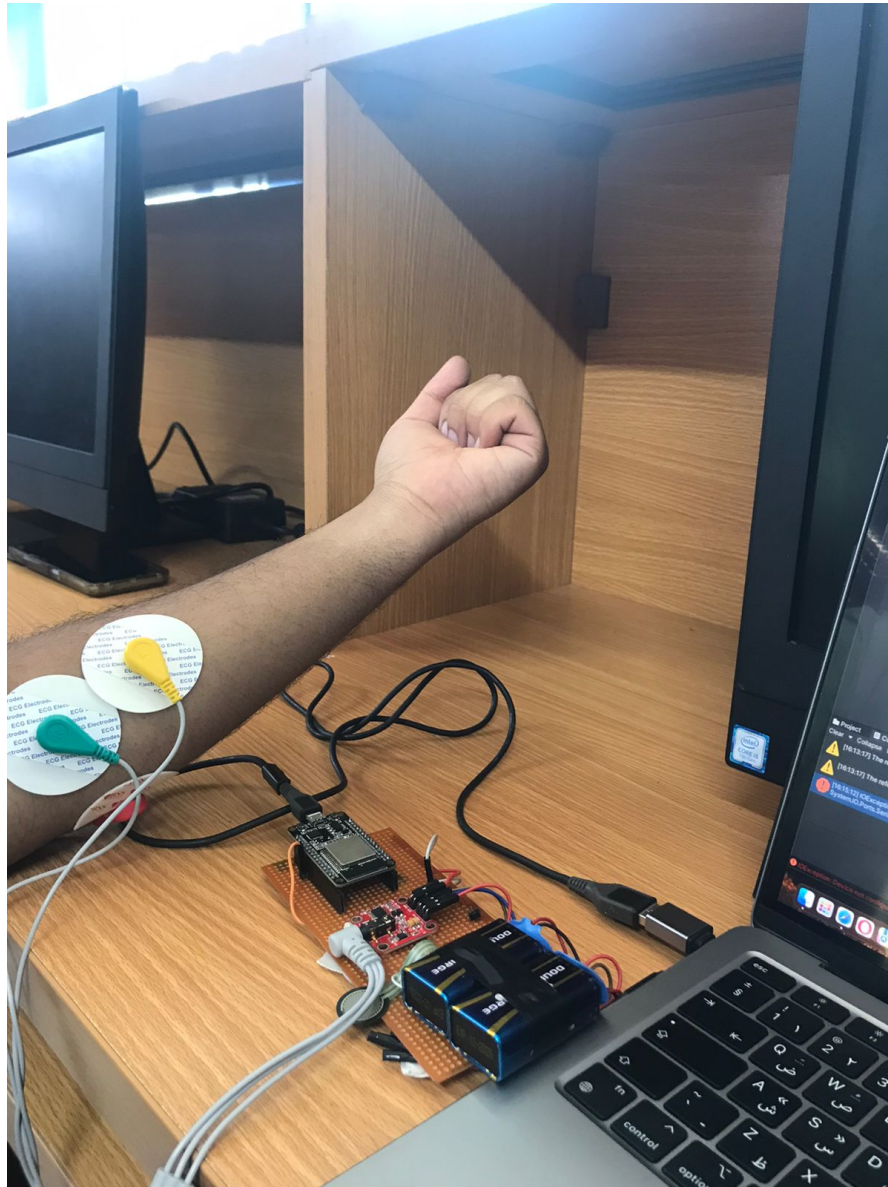
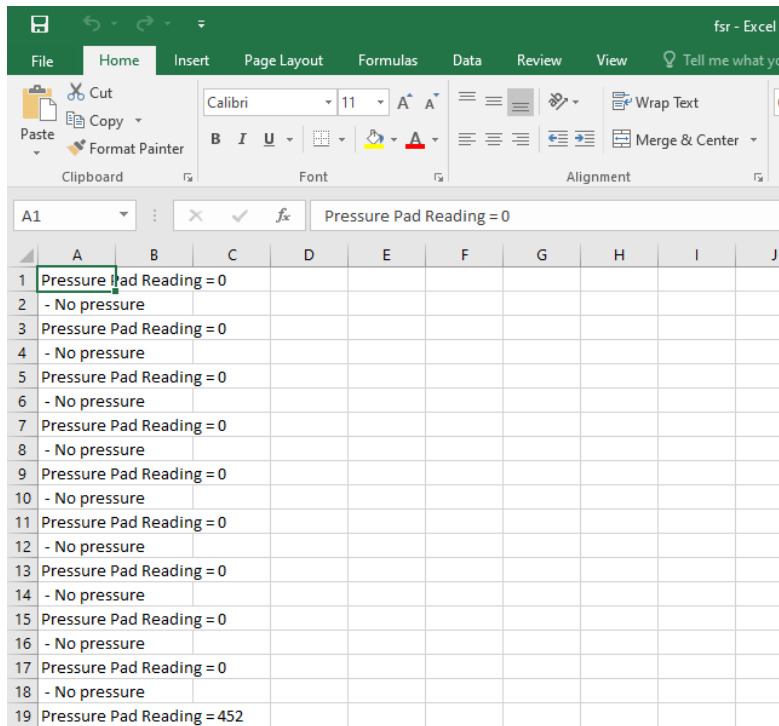


Figure 3.5: Experimental setup for HMI technology

3.5 Software Components

In our project, we will utilize a range of software tools to facilitate various tasks. To begin with, Microsoft Excel, as depicted in Figure 3.6, will serve as our primary tool for data collection and organization. For programming the ESP32 board, we will rely on the Arduino IDE(Figure 3.7). Additionally, we will leverage Google Collabs, as referenced in Figure 3.8, for data filtering and the application of machine learning algorithms. Furthermore, we will incorporate Unity(Figure 3.9), a platform that enables us to control and manipulate avatars within virtual reality environments. This integration with Unity will be instrumental in demonstrating our project's output in a visually immersive and interactive manner.



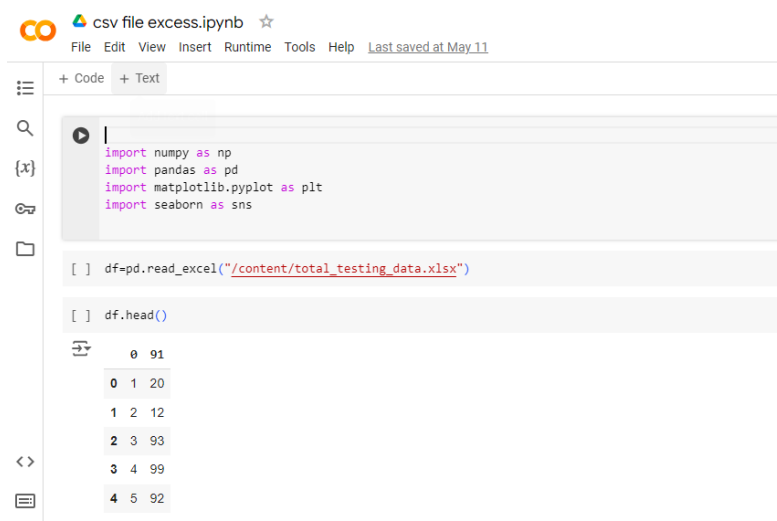
	A	B	C	D	E	F	G	H	I	J
1	Pressure Pad Reading = 0									
2	- No pressure									
3	Pressure Pad Reading = 0									
4	- No pressure									
5	Pressure Pad Reading = 0									
6	- No pressure									
7	Pressure Pad Reading = 0									
8	- No pressure									
9	Pressure Pad Reading = 0									
10	- No pressure									
11	Pressure Pad Reading = 0									
12	- No pressure									
13	Pressure Pad Reading = 0									
14	- No pressure									
15	Pressure Pad Reading = 0									
16	- No pressure									
17	Pressure Pad Reading = 0									
18	- No pressure									
19	Pressure Pad Reading = 452									

Figure 3.6: Dataset for EMG and FSR classification



```
1 int pressureAnalogPin = 0; //Pin where our pressure pad is located.
2 int pressureReading; // INT variable for storing our reading
3 //Adjust these if required.
4 int noPressure = 5; //max value for no pressure on the pad
5 int lightPressure = 100; //max value for light pressure on the pad
6 int mediumPressure = 200; //max value for medium pressure on the pad
7 void setup(void) {
8   Serial.begin(9600);
9 }
10 void loop(void) {
11   pressureReading = analogRead(pressureAnalogPin);
12   Serial.print("Pressure Pad Reading = ");
13   Serial.println(pressureReading);
14   //float voltage = pressureReading * (5.0 / 1023.0);
15   //Serial.print("Voltage = ");
16   //Serial.println(voltage);
17   if (pressureReading < noPressure) {
18     Serial.println(" - No pressure");
19   } else if (pressureReading < lightPressure) {
20     Serial.println(" - Light Pressure");
21   } else if (pressureReading < mediumPressure) {
22     Serial.println(" - Medium Pressure");
23   } else{
24     Serial.println(" - High Pressure");
25   }
26   delay(1000);
27 }
28
```

Figure 3.7: Arduino IDE



```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

[ ] df=pd.read_excel("/content/total_testing_data.xlsx")

[ ] df.head()
```

	0	1	2	3	4	5	91
0	1	20					
1	2	12					
2	3	93					
3	4	99					
4	5	92					

Figure 3.8: Google co-laboratory used for Machine Learning

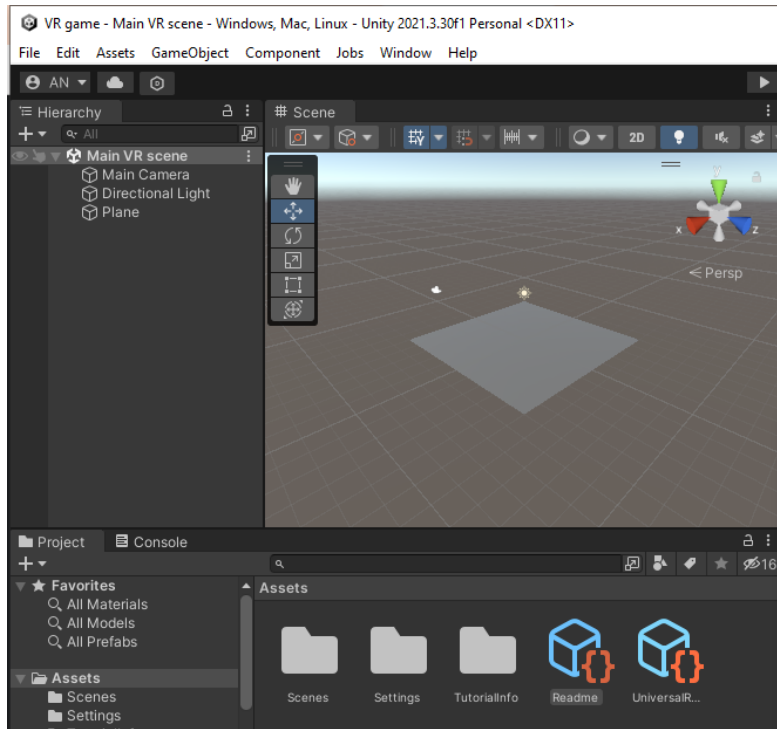


Figure 3.9: Unity

3.6 Results

We used Arduino IDE to simulate muscle signals in the form of waves. To make these waves accurate, we filtered out noise using a band-low pass filter and wavelet transformation. In Figure 3.10, you can see a wave that represents the muscle signal when the hand moves. If the hand is open, the wave is at rest, and if the hand is closed, the wave goes upward. Figure 3.11 shows a table with values from a force-resistive pad, telling us about light, medium, and hard pressure on the finger. In Figure 3.12 we plot the data together of both sensors in the same plane. In Figure 3.13, there's a picture of an excavator avatar that we made in Unity. This avatar corresponds to the muscle signals and forces resistive pad values, giving us a clear picture of how everything works in the excavator simulator. We used Arduino IDE, Python, and Unity to create a smooth system that simulates and shows the muscle signals and pressure applied when operating an excavator. The signals from the EMG sensor and the pressure values from the Force-Sensing Resistors (FSRs) were collected and stored in CSV files. These raw sensor values were then processed by a machine learning algorithm, which classified them into two categories (0s and 1s). The binary signals were subsequently transmitted to Unity for further action. The behavior of the excavator arm was determined by the classified signals. If both the EMG and FSR signals were low, the excavator's arm remained in its original position. When both signals were high, the arm moved downward and the bucket performed a digging action. If the EMG signal was high and the FSR signal was low, the arm moved downward while the bucket stayed in its original position. Conversely, if the EMG signal was low and the FSR signal was high, the arm stayed in its original position while the bucket executed the digging action. This classification and control mechanism ensured precise and context-sensitive operations of the excavator simulator.

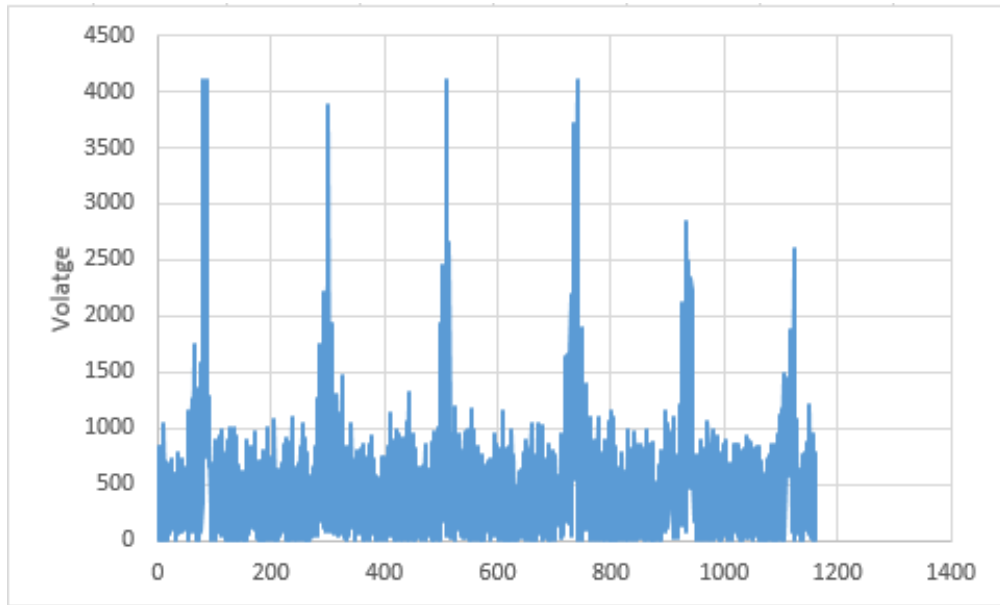


Figure 3.10: Muscles Signals (open and close hand) Wave

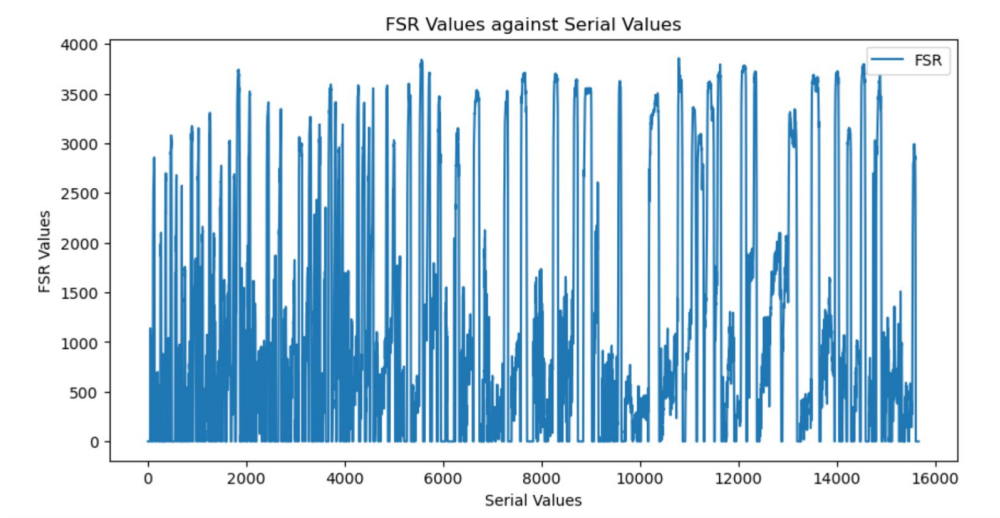


Figure 3.11: Force Resistive Pad Wave

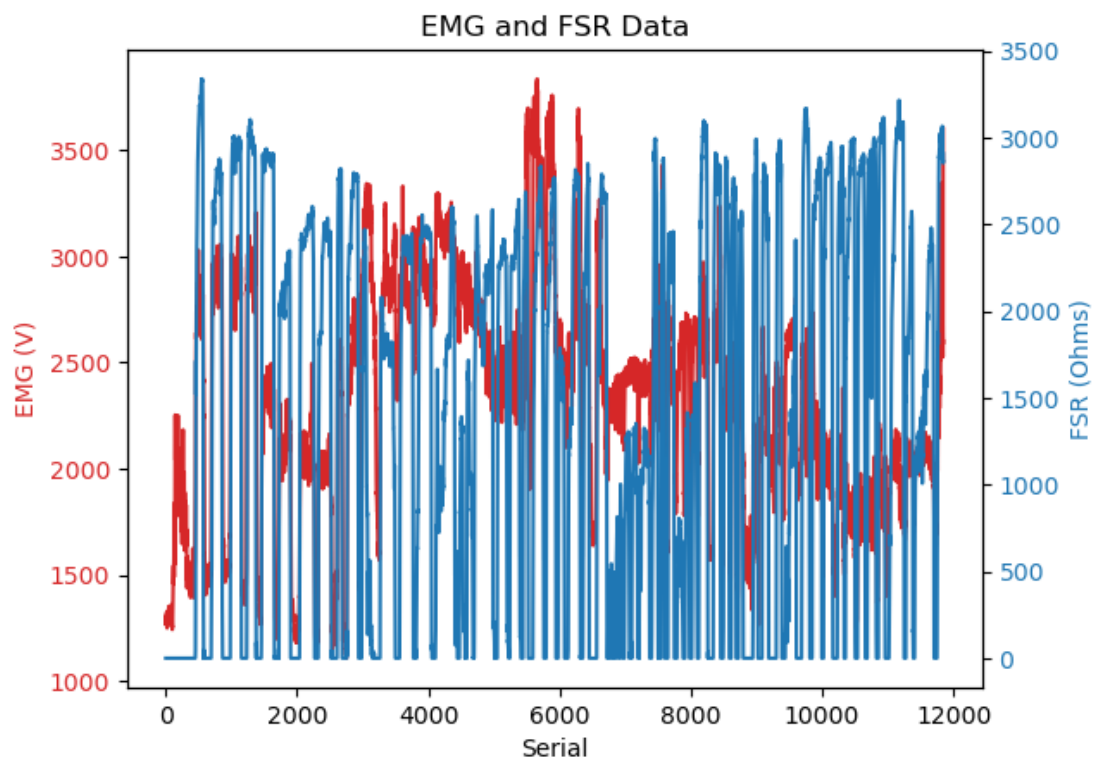


Figure 3.12: EMG and FSR Data wave



Figure 3.13: Excavator Avatar

3.6.1 Convolutional Neural Networks (CNN)

Convolutional Neural Networks (CNNs) are a type of artificial neural network commonly used for analyzing image and video data, but they can also be effectively applied to numerical data, such as time series [19]. In a CNN, the input data is processed through multiple layers of neurons known as convolutional layers. Each convolutional layer applies a set of learnable filters to the input data. These filters slide systematically over the data, performing convolution operations to produce feature maps. For numerical data, such as time series, the filters in the convolutional layers are designed to detect specific patterns or features, such as trends, periodicities, or anomalies, in the data [20]. Each filter captures different aspects of the data, and the resulting feature maps represent the output of the convolutional layer. These feature maps are then used as input for subsequent layers in the network. During the training process, the CNN learns which filters are most effective for detecting relevant features in the numerical data by adjusting the filter weights to minimize the error between the predicted output and the actual output. This allows the network to automatically discover important patterns in the data that are useful for tasks such as sequence prediction or anomaly detection. When applied to time series data, a CNN can learn temporal patterns and relationships between data points over time. By using appropriately designed layers, the network can better capture these temporal patterns, improving its ability to make accurate predictions or detect anomalies in the data.

3.6.2 Layer-wise Convolutional Architecture Optimization

Altering the architecture of convolutional layers in a neural network, such as changing the number of layers, the size of the filters, or the number of filters, can significantly impact the model's performance. By adjusting these parameters, researchers can influence the model's ability to capture different levels of abstraction within the data. Increasing the number of convolutional layers enhances the model's capacity to learn and represent complex patterns, enabling it to identify intricate features and relationships within the data. Furthermore, modifying the filter size and count allows for a more tailored feature extraction process, which can improve the model's ability to recognize and classify patterns effectively. These changes in the convolutional layers are crucial for optimizing the neural network's performance in various tasks, such as image recognition, time series prediction, or anomaly detection. By systematically exploring and implementing these adjustments, researchers can develop more robust and accurate models, capable of handling a wide range of data-driven challenges.

3.7 Results Analysis

The results of the EMG and FSR sensor data are analyzed using three different configurations of a convolutional neural network (CNN) model. Each configuration varies in the number of layers and neurons, impacting their performance metrics such as accuracy and loss.

For EMG, the first configuration (deep learning Configuration 1), which consists of a single layer with 32 neurons, the model achieved an accuracy of 69.01% . This configuration highlights the capability of even a simple network to capture relevant features in the data (Figure 3.14).

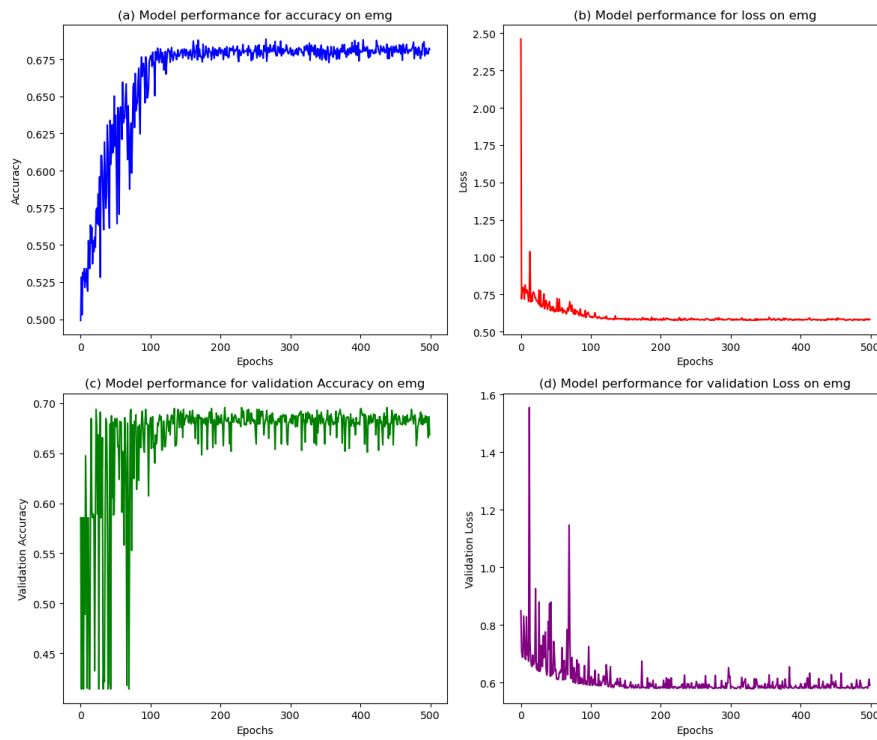


Figure 3.14: EMG Performance Metrics for Deep Learning Configuration 1 (1 Layer, 32 Neurons)

In the second configuration (deep learning Configuration 2), the model was expanded to two layers with 64 neurons. This deeper network demonstrated a significant improvement in performance, achieving an accuracy of 69.46% . This configuration showcases the advantage of increasing the network depth for better feature extraction and pattern recognition (Figure 3.15).

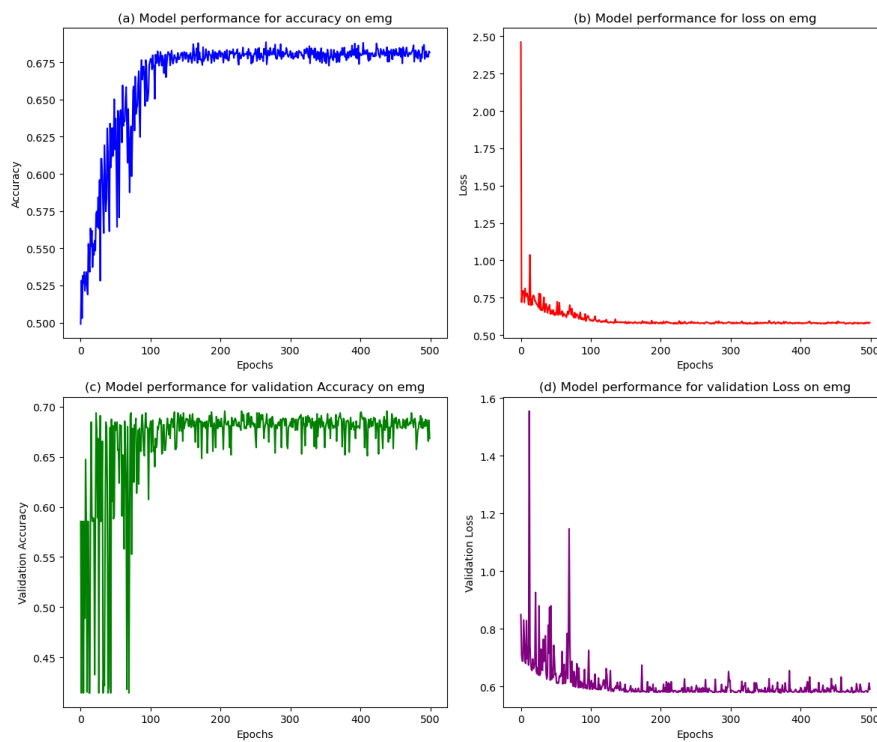


Figure 3.15: EMG Performance Metrics for Deep Learning Configuration 2 (2 Layers, 64 Neurons)

In the third configuration (deep learning Configuration 3), the model was expanded to three layers with 128 neurons. This deeper network demonstrated a significant improvement in performance, achieving an accuracy of 71.02% . This configuration showcases the advantage of increasing the network depth for better feature extraction and pattern recognition (Figure 3.16)

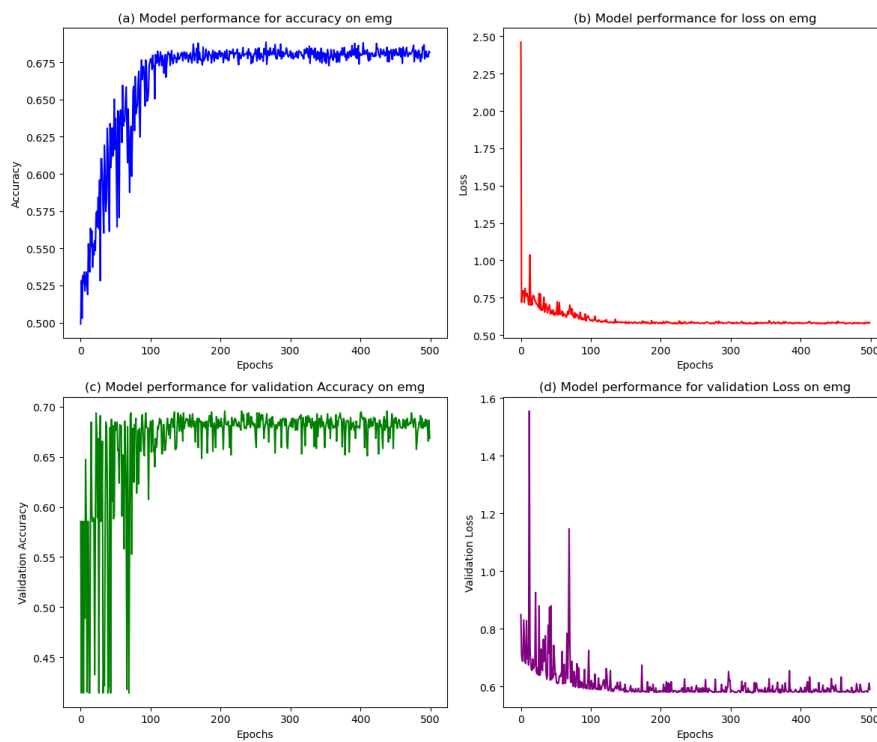


Figure 3.16: EMG Performance Metrics for Deep Learning Configuration 3 (3 Layers, 128 Neurons)

The summary table (Figure 3.17) compares the three configurations by listing the number of layers, neurons, and the count of trainable and non-trainable parameters. This summary underscores how variations in the network's architecture can influence its learning capacity and computational complexity.

NO.	DL Configuration	No. layers	NO. Neurons	Trainable /Non-trainable Parameters
1	Configuration 1	1	32	8833/0
2	Configuration 2	2	64	23233/0
3	Configuration 3	3	128	64321/0

Figure 3.17: EMG Summary of Different Deep Learning Configurations

A comparative analysis of the performance metrics (Figure 3.18) reveals that deep learning Configuration 3 outperforms the others, achieving the highest accuracy and lowest loss.

Deep Learning Configuration	Accuracy (%)	Loss
Configuration 1	69.01	0.5698
Configuration 2	69.46	0.5517
Configuration 3	71.02	0.5436

Figure 3.18: EMG Comparative Performance Metrics of Deep Learning Configurations

The results of the FSR pad data for the first configuration (deep learning Configuration 1), which consists of a single layer with 32 neurons, the model achieved an accuracy of 97.56% . This configuration highlights the capability of even a simple network to capture relevant features in the data (Figure 3.19).

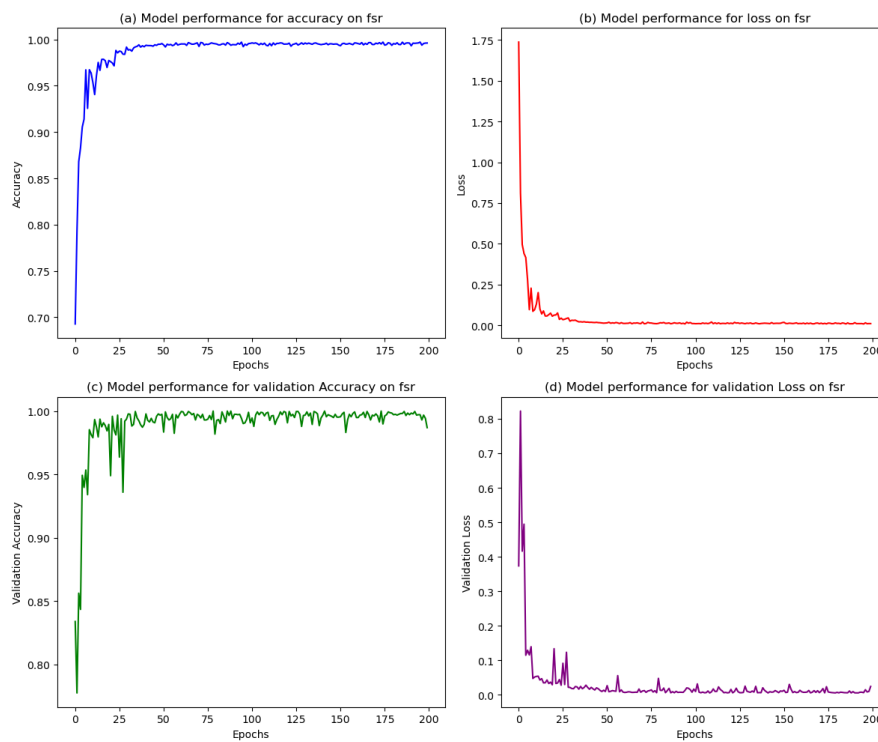


Figure 3.19: FSR Performance Metrics for Deep Learning Configuration 1 (1 Layer, 32 Neurons)

In the second configuration (deep learning Configuration 2), the model was expanded to two layers with 64 neurons. This deeper network demonstrated a significant performance improvement, achieving an accuracy of 99.61%. This configuration showcases the advantage of increasing the network depth for better feature extraction and pattern recognition (Figure 3.20).

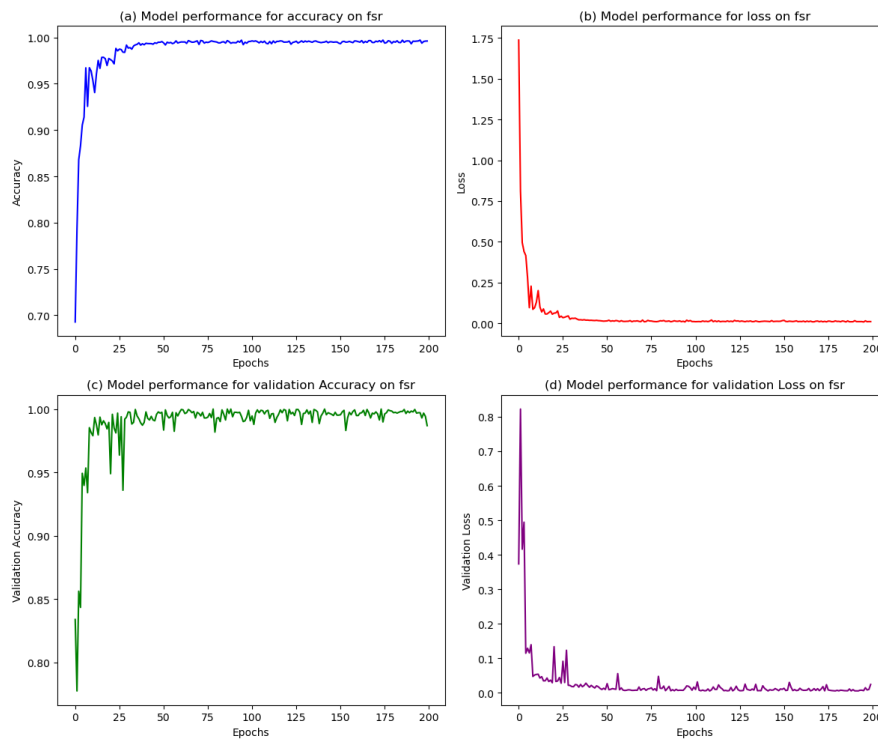


Figure 3.20: FSR Performance Metrics for Deep Learning Configuration 2 (2 Layers, 64 Neurons)

In the third configuration (deep learning Configuration 3), the model was expanded to two layers with 128 neurons. This deeper network demonstrated a significant performance improvement, achieving an accuracy of 99.59%. This configuration showcases the advantage of increasing the network depth for better feature extraction and pattern recognition (Figure 3.21).

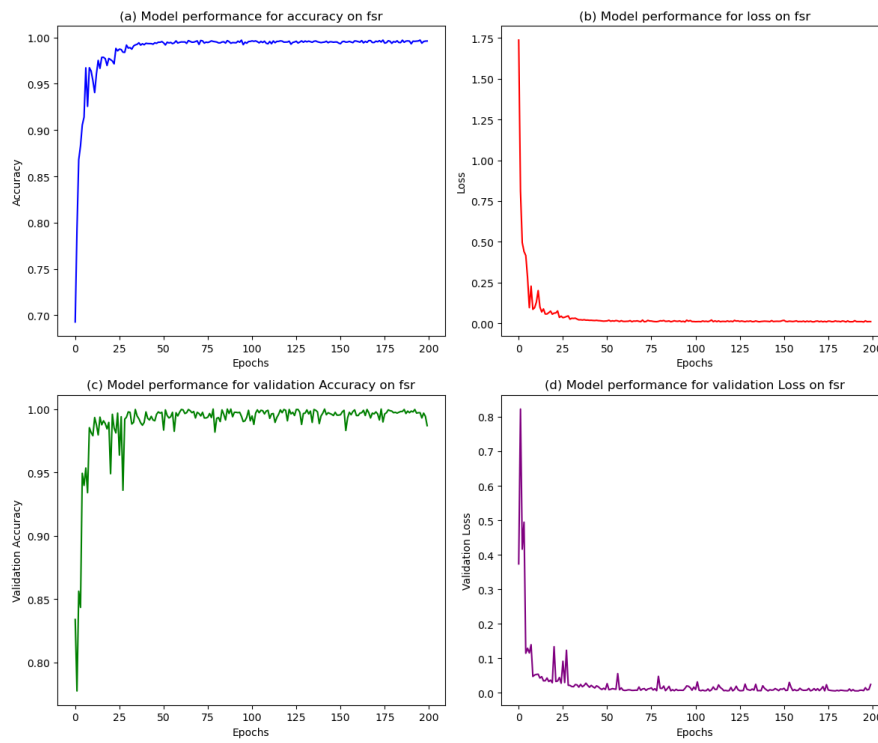


Figure 3.21: FSR Performance Metrics for Deep Learning Configuration 3 (3 Layers, 128 Neurons)

The summary table (Figure 3.22) compares the three configurations by listing the number of layers, neurons, and the count of trainable and non-trainable parameters. This summary underscores how variations in the network's architecture can influence its learning capacity and computational complexity.

No.	Deep Learning configuration	No. of layers	No. of Neurons	Trainable /Non-trainable parameters
1	configuration 1	1	32	9347/0
2	configuration 2	2	64	23233/0
3	configuration 3	3	128	64835/0

Figure 3.22: : FSR Summary of Deep Learning configurations

A comparative analysis of the performance metrics (Figure 3.23) reveals that deep learning Configuration 2 outperforms the others, achieving the highest accuracy and lowest loss.

Deep Learning Configuration	Accuracy (%)	Loss
Configuration 1	99.56	0.0109
Configuration 2	99.61	0.0103
Configuration 3	99.59	0.0117

Figure 3.23: FSR Comparative Performance Metrics of Deep Learning Configurations

In evaluating the efficiency of our sensor fusion system, we meticulously analyzed the individual accuracy rates of our primary sensors: electromyography (EMG) and force-sensitive resistors (FSR). The EMG sensor, renowned for its ability to detect muscular electrical activity, exhibited an accuracy of 71%. Conversely, the FSR sensor, engineered to measure pressure or force exerted on its surface, displayed a notably higher accuracy level of 98%. Furthermore, to provide a comprehensive assessment of our system's performance, we derived the average accuracy by amalgamating the individual accuracy rates of the EMG and FSR sensors and we achieved the accuracy of 84.5% which can be seen in the figure 3.24.

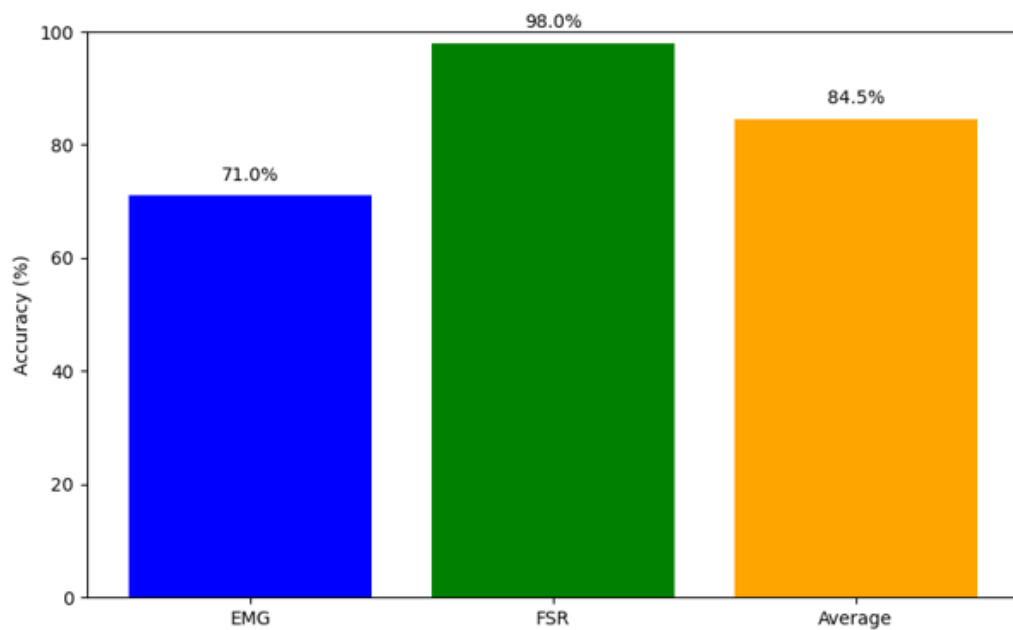


Figure 3.24: Accuracy Comparison of EMG and FSR

3.8 Summary

This chapter provides a comprehensive overview of our proposed solution to address the identified issues. Within this section, we delve into every aspect of our solution. Firstly, we present the core details of our solution, elucidating its functionality and highlighting its advantages. Secondly, we share the packaging concept for the prototype, emphasizing its compact and dependable design to enhance portability and accessibility. The third section discusses the key components that will be incorporated into our project to serve various purposes. The fourth part offers intricate insights into the software applications utilized in our project. This chapter encompasses all facets of our project, from its fundamental implementation to the programming and the user interface, offering a thorough explanation of each element.

CHAPTER 4

MILESTONES, WORK DIVISION AND COST

4.1 Milestones

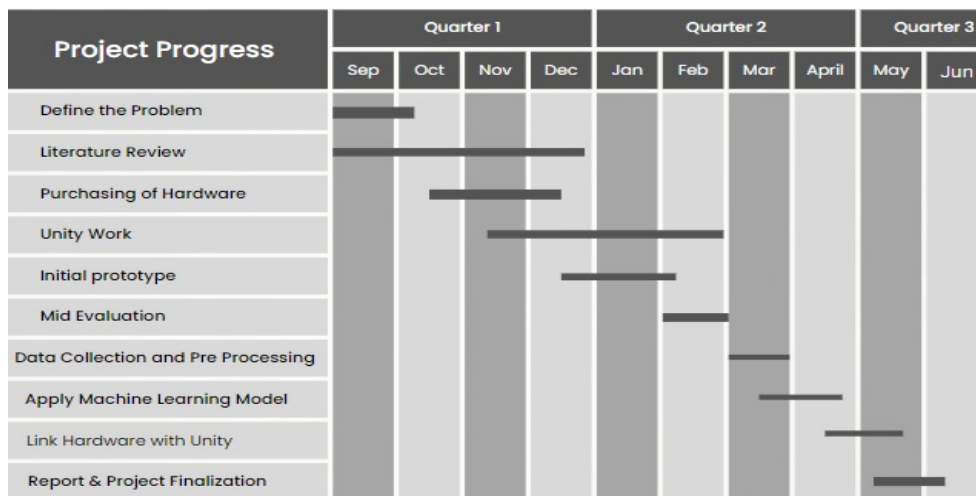


Figure 4.1: Timeline

4.2 Work Division

Month	Task distribution among group members
September 2023	Amna Tanveer: Read at least five research papers on the EMG sensor module. Ansa Niaz: Read at least five research papers on the EMG sensor module and unity. Abdullah Bin Omar Jawaaid: Read at least five research papers on the EMG sensor module and FSR.
October 2023	Amna Tanveer: Hardware implementation of Arduino and EMG sensor and collection of some data sets. Ansa Niaz: Work on unity and follow the unity tutorials. Abdullah Bin Omar Jawaaid: Hardware implementation of Arduino and EMG sensor and collection of some data sets.
November 2023	Amna Tanveer: Write the FYP report part before the FYP defence and overall finalize it on the IEEE pattern using Latex. Ansa Niaz: Write the FYP report part before the FYP defence and overall finalize it on the IEEE pattern using Latex. Abdullah Bin Omar Jawaaid: Make the presentation before the FYP defence.
December 2023	Amna Tanveer: Working on the wave and trying to remove the noise signals by using low-pass filters. Ansa Niaz: Training and moving avatar in unity. Abdullah Bin Omar Jawaaid: Working on the wave and trying to remove the noise signals by using low-pass filters.
January 2024	Amna Tanveer: Write the FYP report part before the FYP mid-defense and overall finalize it on the IEEE pattern using Latex and trying to work on force-sensitive resistive pad. Ansa Niaz: Write the FYP report part before the FYP defence and overall finalize it on the IEEE pattern using Latex and try to make an avatar of the excavator on unity. Abdullah Bin Omar Jawaaid: Make the presentation before the FYP mid-defence and try to work on the force-sensitive resistive pad.
February 2024	Amna Tanveer: Working on the wave of EMG and FSR. Ansa Niaz: Making and moving excavator in unity. Abdullah Bin Omar Jawaaid: Working on the wave of EMG and FSR.
March 2024	Amna Tanveer: Working on the wave of both sensors and collecting the dataset of both sensors. Ansa Niaz: Training and moving the avatar in unity by reading the values CSV files. Abdullah Bin Omar Jawaaid: Working on the wave of both sensors and collecting the dataset of both sensors.
April 2024	Amna Tanveer: Working on the wave of both sensors trying to run in the same microcontroller at the same time. Ansa Niaz: Training to connect the unity with the hardware. Abdullah Bin Omar Jawaaid: Working on the wave of both sensors trying to connect both sensors to one Microcontroller.

CHAPTER 5

CONCLUSION

5.1 Conclusion

To sum up, this dissertation describes our path toward creating a solution based on sEMG technology. Our main objective was to provide a workable, domestically produced, and widely accepted solution to the problems related to the interpretation of muscle signals. To create an effective and useful result, we worked through the project by utilizing a variety of tools, approaches, software programs, and algorithms. Utilizing EMG sensor modules and force-sensitive resistive pad(FSR), ESP32 for wireless connection, and Unity software to create an immersive virtual environment are some of the components of our approach. Processing and replicating muscle signals was made easier by the Arduino IDE. In addition, Convolutional Neural Networks (CNN) models were used as part of our machine learning methodologies to train and analyze the gathered data. We were able to improve the precision and effectiveness of our solution by using this method. The determination and collaboration of our group were essential to overcoming obstacles. The process's highs and lows were viewed as teaching moments that helped our project get better over time. Comparable to plunging into a sea of knowledge, the research phase allowed us to comprehend contemporary issues and investigate creative solutions from a variety of sources, including online comments and evaluations. Understanding our solution's composition, unique characteristics, and appearance is made easier by reading the thorough description. Affordability is emphasized, ensuring usefulness for a larger group of people. Moreover, the development of the sEMG-based excavator simulator marks a significant milestone in the realm of virtual reality training for heavy machinery operators. By harnessing wearable surface electromyography (sEMG) technology and integrating force-sensing resistors (FSRs), our simulator not only provides a cost-effective alternative to traditional training methods but also offers an immersive and realistic training experience that mirrors actual excavator operations. Despite encountering challenges in obtaining accurate muscle signal data and ensuring real-time model precision, our project effectively addresses critical needs in the construction and heavy machinery industry. Beyond merely enhancing operator skills and safety, our simulator contributes to substantial cost savings and environmental sustainability by reducing the dependence on actual machinery for training purposes. In the future, we want to build on this project and consider incorporating new tools, approaches, and innovations to make our solution even more advanced and significant. We want to upgrade our project by adopting multi-channel surface electromyography (sEMG) data acquisition, allowing for more precise monitoring of muscle activity.

5.2 Future Work

In the future, we aim to enhance our project by integrating advanced functionalities, starting with the adoption of multi-channel surface electromyography (sEMG) data acquisition. Currently, our setup uses a single-channel configuration with three leads to capture muscle signals. By transitioning to an eight-channel setup, we can significantly enhance the granularity of our data acquisition, allowing for more precise monitoring of muscle activity. This shift will enable us to capture detailed signals from various body regions, utilizing full-body EMG sensors. This expansion not only broadens the scope of our project but also opens up possibilities for a wider range of immersive VR experiences. However, increasing the number of channels introduces hardware constraints, and future work will focus on optimizing the system to operate efficiently within these practical limitations. As our project evolves in complexity, we plan to employ transform theorems to streamline the data processing and analysis, ensuring efficient and accurate interpretation of muscle signals. Additionally, we aim to explore the incorporation of a joystick as an alternative control method alongside sEMG. This would offer users a familiar control scheme while potentially simplifying complex maneuvers. By strategically integrating these advancements, we aim to fortify the effectiveness and reliability of our project, paving the way for more immersive and precise VR training solutions.

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